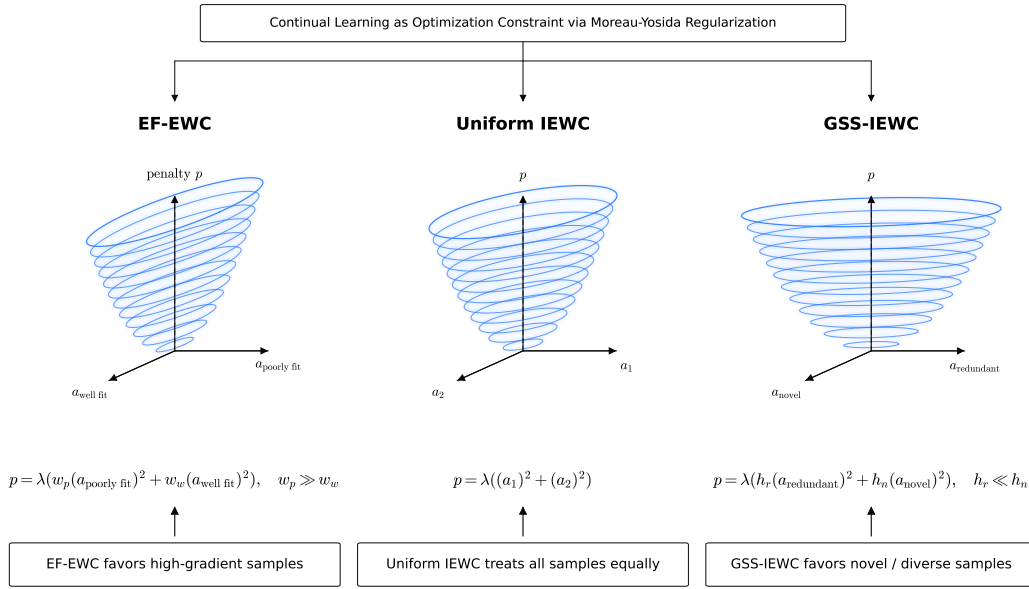


# Improved Elastic Weight Consolidation as an Optimization Constraint for Continual Learning

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**Figure 1:** Each surface shows the continual-learning penalty over normalized directions, which locally approximate loss changes for two representative samples. EF-EWC induces anisotropy through gradient-scale weighting; uniform IEWC removes this scaling, while GSS-IEWC weights samples by diversity. [1]

## Abstract

Elastic Weight Consolidation (EWC) is widely used in continual learning, and standard implementations use the empirical Fisher matrix, whose justification beyond probabilistic classification remains incomplete. The empirical Fisher differs from the true Fisher, aligns with it only under specific assumptions, and is nevertheless used broadly in practice. We address this gap by deriving EWC from first principles. Starting from the continual-learning problem, we formulate old-task solution preservation as a constrained optimization problem (CLOC) and develop PL-CLOC, a population, low-order variant for deriving scalable mechanisms. Within this framework, preserving the old scalar loss value recovers exactly the empirical Fisher matrix (EF) used in EWC. This places empirical-Fisher EWC on direct optimization-theoretic footing without reliance on the true Fisher matrix. Preserving outputs up to motion along the old-task loss level set recovers the improved empirical Fisher (IEF) and leads to IEWC, a generalized and normalized sibling of EWC. EF implicitly scales local old-task sensitivity directions by the output’s gradient magnitude, whereas IEF does not. We identify EWC-DR’s vanishing-importance pathology as a special case of this gradient-scale effect on well-fitted data, and

provide the improved empirical Fisher as the general solution. More broadly, the derivation separates the outer EWC mechanism from output geometry, enabling geometry-aware continual learning within the same regularization template. Empirically, across image classification, continual LLM/NLP, segmentation, forecasting, and diffusion, IEWC improves final average performance in two settings and, with sample weighting, improves performance across all tested settings. Together, these results motivate IEWC as a theoretical and practical successor.

**Code:** <https://github.com/Axym-Labs/iewc>

**Project page:** <https://axym.org/work/iewc>

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# 1 Continual Learning as Optimization Constraint

Consider the problem of continual learning, where data from distributions (tasks, environments) are presented sequentially [2–5]. Concretely, we first train a model on distribution  $A$  and obtain parameters  $\theta_A^*$ , and only then can we optimize with respect to a new distribution  $B$ . If we plainly pick a minimal loss/cost  $\min_{\theta} L_B(\theta)$  as our objective, then the optimization does not have to respect performance on  $A$ , which usually degrades it too much to be practical. Hence, we have to guide or constrain optimization in a way. A natural fit for this problem is the framework of constrained optimization, with solution preservation being the constraint in question. Define both a quantity  $Q : \Theta \rightarrow \mathcal{H}$  that is relevant to this preservation and a constraint set  $\mathcal{C} \subset \mathcal{H}$  that implies preservation is met. Before exploring concrete choices in later sections, we continue on the common thread. These two objects let us define a new objective:

$$\min_{\theta \in \Theta} L_B(\theta) \quad \text{subject to} \quad q(\theta) \in \mathcal{C}$$

This gives a mathematical formulation of preservation, and the next section extends it to become more practical.

## 2 Population and Local Approximations

To obtain a differentiable objective suitable for gradient-based optimization, we apply the standard Moreau-Yosida regularization of the constraint indicator, which for closed convex  $\mathcal{C}$  yields the squared-distance penalty [6]:

$$\min_{\theta} J(\theta) = \min_{\theta} L_B(\theta) + \frac{\lambda}{2} \text{dist}(q(\theta), \mathcal{C})^2 = \min_{\theta} L_B(\theta) + \frac{\lambda}{2} \inf_{u \in \mathcal{C}} \text{dist}(q(\theta), u)^2.$$

We call this the **Continual Learning as Optimization Constraint (CLOC)** formulation. GEM and A-GEM are similar in spirit in that they formulate preservation through constraints on old examples [7, 8]. CLOC turns the old-task constraint into a proximal penalty built after task- $A$  training. In a population setting, e.g. when we have a dataset, we can translate this into a population risk objective by wrapping it in an expectation over the data and using sample-based analogues  $\ell_B$  of  $L_B$  and  $q$  of  $Q$ , then taking the empirical average. Define the unscaled proximal loss as  $p(a, \theta) = \text{dist}(q(a, \theta), \mathcal{C})^2$ .

$$\min_{\theta} \frac{1}{N_B} \sum_{i=1}^{N_B} \ell_B(b_i, \theta) + \frac{\lambda}{2N_A} \sum_{i=1}^{N_A} p(a_i, \theta)$$

The two-distribution presentation is only for notational simplicity; additional old distributions can be handled by adding analogous constraints. We refer to this population version as **population CLOC (P-CLOC)**. One efficiency consideration is the penalty  $p$ : Ideally, we let the constraint contribute to each gradient step with its gradient. However, recomputing the full old-distribution penalty  $p$  and its associated factors many times during optimization on distribution  $B$  is usually prohibitively expensive, since it depends on the dataset  $D_A$  associated with  $A$ . Additionally, reusing dataset  $D_A$  later on does not respect the continual learning setting. For this reason, replace the exact penalty by a local  $k$ -th order Taylor-polynomial approximation around the old solution  $\theta_A^*$ . This turns the computation of  $p$  into a repeated evaluation of precomputed derivative tensors and bounds memory requirements to  $O(d^k)$  for a  $d$ -dimensional parameterization. Since memory scales exponentially with  $k$ , the standard scalable surrogate is a low-order Taylor approximation. We refer to this local approximation as **low-order CLOC (L-CLOC)**, and to its population version as **PL-CLOC**. It remains to specify the preserved quantity  $q$  and constraint set  $\mathcal{C}$ . Notice that the gradient of  $p$  is dependent on only one sample at once, which allows us to analyze  $p$  and its gradients on a per-sample basis.

### 3 Old-Task Loss Preservation

Since  $\ell_A$  is our best practical surrogate on performance on distribution  $A$ , a natural choice is to use  $\ell_A$  itself to define preservation. We say that the model  $f_\theta$  preserved the solution if it preserved the old loss value for distribution  $A$ . This leads to:

$$q(a, \theta) = \ell_A(f_\theta(x)) - \ell_A(f_{\theta_A^*}(x)).$$

Since preservation means  $q(a, \theta) = 0$ , take  $\mathcal{C} = \{0\}$ . The corresponding unscaled proximal loss is

$$p(a, \theta) = \text{dist}(q(a, \theta), \mathcal{C})^2 = q(a, \theta)^2.$$

To obtain a low-order Taylor approximation, use the perturbation around the old solution  $\delta = \theta - \theta_A^*$  in the first-order expansion of  $q$ :

$$q(a, \theta_A^* + \delta) \approx g^\top \delta,$$

where

$$g = \nabla_\theta \ell_A(f_{\theta_A^*}(a)).$$

Equivalently, writing

$$z_A = f_{\theta_A^*}(a) \quad r_i = \nabla_z \ell_A(z)|_{z=f_{\theta_A^*}(a)}, \quad J_i = \nabla_\theta f_{\theta_A^*}(a),$$

the chain rule gives

$$g_i = J_i^\top r_i.$$

Substituting this expansion into the preservation penalty gives

$$\frac{\lambda}{2N_A} \sum_{i=1}^{N_A} q(a_i, \theta)^2 \approx \frac{\lambda}{2N_A} \sum_{i=1}^{N_A} (g_i^\top \delta)^2.$$

Collecting the gradient outer products yields

$$\frac{\lambda}{2N_A} \sum_{i=1}^{N_A} (g_i^\top \delta)^2 = \frac{\lambda}{2} \delta^\top \left( \frac{1}{N_A} \sum_{i=1}^{N_A} g_i g_i^\top \right) \delta = \frac{\lambda}{2} \delta^\top M_1 \delta.$$

Therefore the resulting matrix is

$$M_1 = \frac{1}{N_A} \sum_{i=1}^{N_A} g_i g_i^\top = \frac{1}{N_A} \sum_{i=1}^{N_A} J_i^\top r_i r_i^\top J_i.$$

We call this an importance matrix as it encodes how important it is to preserve a direction in parameter space.

### 4 Empirical Fisher Regularization

This is exactly the empirical Fisher matrix  $F_{\text{EF}}$ , the standard practical matrix used in Elastic Weight Consolidation (EWC) [9]. We refer to this instantiation as EF-EWC:

$$M_1 = F_{\text{EF}}.$$

Hence EF-EWC approximates a constrained optimization problem with a smooth proximal penalty, preserving the old scalar loss value sample by sample. This places EF-EWC within the broader family of parameter-importance regularizers [10, 11] and Bayesian or online continual-learning approximations [12–14].

The original EF-EWC derivation uses properties specific to probabilistic classification models. After training on task  $A$ , one considers the posterior over parameters

$$p(\theta \mid \mathcal{D}_A) \propto p(\mathcal{D}_A \mid \theta) p(\theta),$$

and uses it as a prior when learning task  $B$ . Expanding the negative log-posterior around the old solution  $\theta_A^*$  yields a local quadratic approximation

$$-\log p(\theta \mid \mathcal{D}_A) \approx -\log p(\theta_A^* \mid \mathcal{D}_A) + \frac{1}{2}(\theta - \theta_A^*)^\top H_A(\theta_A^*)(\theta - \theta_A^*),$$

where  $H_A(\theta_A^*)$  is the Hessian of the negative log-posterior at the old solution. Under the usual assumptions for log-likelihood models, and neglecting the prior curvature term or absorbing it into the approximation, this Hessian can be replaced by the Fisher information matrix

$$F(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}_A} \left[ \nabla_\theta \log p_\theta(y \mid x) \nabla_\theta \log p_\theta(y \mid x)^\top \right].$$

For negative log-likelihood losses, this Fisher matrix coincides with the curvature of the expected loss under standard regularity conditions, and therefore matches the corresponding Hessian-based local quadratic model near an optimum, as in the natural-gradient and approximate-curvature literature [15–17]. Since classification models explicitly define a predictive distribution  $p_\theta(y \mid x)$  over labels, the true Fisher can be computed as population object by sampling from said distribution. In practice, however, EWC replaces it by empirical approximations computed from the old dataset, which leads to the empirical Fisher form used above. However, this framework leaves open a justification of the empirical Fisher outside of negative log-likelihood losses. This is an open problem because EWC is used for different tasks and objectives as well, for example for diffusion [18]. Furthermore, Kunstner et al. show that the empirical Fisher is a different mathematical object than the true Fisher and approximates the true Fisher well only in specific settings [19]. We re-derive EF-EWC from scratch using PL-CLOC to resolve these two issues, basing the usage of said matrix on solid theoretical grounding without reliance on the true Fisher matrix.

## 5 Level-Set Preservation in Output Space

Our first mechanism is derived from preservation of the loss value. An extension of the same preservation constraint is to impose it through the old-task loss level set in output space. More concretely, we preserve the output per-sample up to motions that stay on this set. This output-space perspective connects to distillation and functional regularization approaches to continual learning [20–23], which let the model match old predictions or predictive functions. This gives a geometric formulation of the same underlying constraint: instead of penalizing raw change in the scalar loss coordinate, we penalize displacement away from the local constraint set in output space. Let  $a \in D_A$  be an old sample, and write  $z_A = f_{\theta_A^*}(x)$ . The exact old-loss preservation set in output coordinates is

$$\mathcal{C} = \{z : \ell_A(z) = \ell_A(z_A)\}.$$

Thus the preserved quantity is the model output itself:

$$q(a, \theta) = f_\theta(x).$$

The hard constraint is

$$q(a, \theta) \in \mathcal{C},$$

and the corresponding unscaled proximal loss is

$$p(a, \theta) = \text{dist}_G(q(a, \theta), \mathcal{C})^2,$$

where  $G \succ 0$  is the chosen local output metric.

This exact penalty is generally not tractable. To obtain a practical form, use the standard first-order local model of the constraint set near the old output  $z_A$ , namely its tangent plane. Let

$$r = \nabla_z \ell_A(z) \big|_{z=z_A}.$$

For a small output displacement  $\Delta z = f_\theta(x) - f_{\theta_A^*}(x)$ , the first-order old-loss change is approximately  $r^\top \Delta z$ . Hence the local constraint set is

$$T = \{\Delta z : r^\top \Delta z = 0\}.$$

Replacing the exact level set by this tangent hyperplane gives the local proximal loss

$$p(a, \theta) \approx \text{dist}_G(\Delta z, T)^2.$$

For the metric  $|v|_G^2 = v^\top G v$ , the squared distance to the tangent hyperplane follows from projecting  $\Delta z$  onto the gradient:

$$\text{dist}_G(\Delta z, T)^2 = \frac{(r^\top \Delta z)^2}{r^\top G^{-1} r}.$$

Linearizing the model output around  $\theta_A^*$ , with

$$\delta = \theta - \theta_A^*, \quad J = \nabla_\theta f_{\theta_A^*}(x),$$

we have

$$\Delta z \approx J \delta.$$

Substituting this into the local regularization term gives

$$\frac{\lambda}{2} p(a, \theta_A^* + \delta) \approx \frac{\lambda}{2} \frac{(r^\top J \delta)^2}{r^\top G^{-1} r}.$$

Equivalently,

$$\frac{\lambda}{2} p(a, \theta_A^* + \delta) \approx \frac{\lambda}{2} \delta^\top \left( J^\top \frac{r r^\top}{r^\top G^{-1} r} J \right) \delta.$$

Taking the empirical average over old-distribution samples  $a_1, \dots, a_N$  yields an importance matrix of the form

$$M_{2,A} = \frac{1}{N} \sum_{i=1}^N J_i^\top \frac{r_i r_i^\top}{r_i^\top G_i^{-1} r_i} J_i.$$

## 6 Improved Empirical Fisher Regularization

In the Eucliden-output csae, this recovers the improved empirical Fisher matrix introduced by Wu et al. [24]:

$$M_{2,I} = F_{\text{IEF}}.$$

Hence the IEF is the first-order local approximation of a constrained optimization problem that protects old behavior geometrically: instead of preserving the scalar loss value, it preserves the output up to motion along the loss level set. The original IEF paper derives the same normalization from approximate natural-gradient descent. In that setting, the empirical Fisher reweights sample contributions by squared output-gradient magnitude, which can make it a poor curvature surrogate; IEF removes this scale factor and retains the normalized output-loss direction. In our derivation, the same normalization arises because the penalty is defined through distance to the old-loss level set rather than through raw change in the scalar loss coordinate.

## 7 Relation to Gauss-Newton

Consider the scalar least-squares setting in Euclidean geometry. Suppose the model output is scalar, such that

$$\ell_n(z) = \frac{1}{2}(z - y_n)^2.$$

Then

$$r_n = z_n^A - y_n$$

is scalar, and for  $r_n \neq 0$  we have

$$\frac{r_n^2}{r_n^2} = 1.$$

Therefore the Euclidean IEF contribution reduces to

$$J_n^\top J_n,$$

and averaging gives

$$M_{\text{IEF}} = \frac{1}{N} \sum_{n=1}^N J_n^\top J_n.$$

This is exactly the Gauss–Newton or output-Jacobian form for squared-error regression [16, 25, 26], the standard practical preconditioning matrix for second-order optimization. This matches our geometric intuition: Loss preservation is governed by how parameters move the output, which is encoded by the Jacobian  $J$ , not by the current output gradient magnitude. This equivalence was made explicit in [24]. In fact, the IEF was motivated as an approximate Gauss–Newton algorithm.

## 8 Normalization and Sample Weighting

To compare both matrices in a common notation, let  $G_n \succ 0$  denote the chosen output metric for sample  $n$ . Although  $F_{\text{EF}}$  itself does not depend on  $G_n$ , its summands can be rewritten relative to said output metric.

Define

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \quad a_n^G = J_n^\top \hat{r}_n^G, \quad w_n^G = r_n^\top G_n^{-1} r_n.$$

Then

$$M_{\text{IEF},G} = \frac{1}{N} \sum_{n=1}^N a_n^G (a_n^G)^\top, \quad F_{\text{EF}} = \frac{1}{N} \sum_{n=1}^N w_n^G a_n^G (a_n^G)^\top.$$

If we interpret both matrices as empirical approximations of local old-task sensitivity directions, represented by  $a_n^G (a_n^G)^\top$ , then they use the same directions but differ in how strongly each sample contributes.  $M_{\text{IEF},G}$  keeps the original empirical weighting of the old-task samples, while  $F_{\text{EF}}$  multiplies each sample’s contribution by the metric-dependent squared magnitude

$$w_n^G = r_n^\top G_n^{-1} r_n.$$

Thus, normalization changes the sample aggregation: EF changes the old-task weighting toward high-loss samples and away from confidently fitted ones, whereas IEF keeps the empirical old-task weighting.

We will analyze this difference more closely below and see that this particular weighting is problematic in general. The next section treats the important special case of probabilistic classification models.

## 9 Gradient-Scale Effects in Classification

The output-gradient weighting in EF has an important classification-specific symptom. Consider one data sample. For cross-entropy with softmax probabilities  $p_n$  and one-hot label  $y_n$ , the output-loss gradient with respect to logits is

$$r_n = p_n - y_n.$$

For a confidently correct prediction with class  $c$  and small  $\varepsilon$ ,

$$p_{n,c} = 1 - \varepsilon,$$

we have

$$\|r_n\|_2^2 = \|p_n - y_n\|_2^2 \in \Theta(\varepsilon^2).$$

Thus the EF contribution of this sample becomes small as the model becomes more confident and correct. This is coherent with the way we derived the EF: the scalar first-order loss change becomes small. In continual learning, this is undesirable: The samples that the old model classifies best may receive very low importance. EWC-DR identifies this failure mode in classification and proposes a classification-specific repair through a logit-reversal operation, literally flipping the signs of the model's logits [27]. In the present framework, their observation can be read as a symptom of the weighting issue identified in Section 8:  $F_{\text{EF}}$  importance depends on output-gradient scale. This evidently does not occur for  $M_{\text{IEF}}$ , and we make this point precise in the next section. Hence, the normalization analysis yields a formal problem identification and its generalizable solution, the Improved Empirical Fisher matrix.

## 10 Behavior Near Well-Fitted Solutions

The loss scale dependence of  $F_{\text{EF}}$  is further accentuated by the fact that we calculate  $F_{\text{EF}}$  *after* training on  $A$ , where we expect the model to have fitted the data well. In fact, this approximate optimality of  $\theta_A^*$  is a necessary assumption in the derivation of EF-EWC [9] to relate the true Fisher to the Hessian. This also matches modern practice, where highly expressive models are often trained to fit the data very well, frequently up to interpolation [28]. This justifies theoretical treatment outside of the classification case.

To make the comparison precise, consider the per-sample summands

$$S_n^{\text{EF}} = J_n^\top r_n r_n^\top J_n, \quad S_n^{\text{IEF},G} = J_n^\top \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n} J_n,$$

where  $G_n \succ 0$  is the chosen output metric. We assume that  $G_n$  is uniformly positive definite and bounded, and that  $J_n$  is bounded under the operator norm. More precisely, assume there exists a constant  $C_J < \infty$  such that

$$\|J_n^\top v\|_2 \leq C_J \|v\|_2$$

for the relevant directions, and constants  $0 < m_G \leq M_G < \infty$  such that  $G_n$  is uniformly bounded between  $m_G I$  and  $M_G I$ .

Using the same  $G_n$ -normalized direction as above,

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \quad a_n^G = J_n^\top \hat{r}_n^G, \quad w_n^G = r_n^\top G_n^{-1} r_n, \quad b_n^G = \|J_n^\top \hat{r}_n^G\|_2^2,$$

This gives

$$S_n^{\text{IEF},G} = a_n^G (a_n^G)^\top, \quad S_n^{\text{EF}} = w_n^G a_n^G (a_n^G)^\top.$$



For the spectral norm,

$$\|S_n^{\text{EF}}\|_2 = w_n^G b_n^G, \quad \|S_n^{\text{IEF},G}\|_2 = b_n^G.$$

Since

$$\|\hat{r}_n^G\|_2^2 = \frac{\|r_n\|_2^2}{r_n^\top G_n^{-1} r_n},$$

the metric bounds and bounded-Jacobian assumption give  $b_n^G \in O(1)$ , so

$$\|S_n^{\text{EF}}\|_2 \in O(w_n^G), \quad \|S_n^{\text{IEF},G}\|_2 \in O(1).$$

The sharper lower-bound statement requires a nondegeneracy condition in the normalized output-gradient direction. If there exists  $c_I > 0$  such that

$$\|J_n^\top \hat{r}_n^G\|_2 \geq c_I$$

for the relevant samples, then

$$\|S_n^{\text{EF}}\|_2 \in \Theta(w_n^G), \quad \|S_n^{\text{IEF},G}\|_2 \in \Theta(1).$$

We now interpret this for practical situations. Near the end of training on distribution  $A$ , we expect many old samples to have small gradients. The norm analysis above then shows that EF contains the additional multiplicative factor  $w_n^G = r_n^\top G_n^{-1} r_n$ , whereas IEF removes that factor. Remaining low-magnitude summands of the IEF come from the model sensitivity term  $b_n^G$ , not from the scale of the output-loss gradient itself. In the classification setting with  $G_n = I$ , a confidently correct prediction with  $p_{n,c} = 1 - \varepsilon$  yields

$$\|r_n\|_2^2 = \|p_n - y_n\|_2^2 \in \Theta(\varepsilon^2).$$

Therefore

$$\|S_n^{\text{EF}}\|_2 = w_n^I b_n^I \in \Theta(\varepsilon^2) b_n^I, \quad \|S_n^{\text{IEF},I}\|_2 = b_n^I.$$

Under the same nondegeneracy condition on  $J_n^\top \hat{r}_n^I$ , this gives  $\|S_n^{\text{EF}}\|_2 \in \Theta(\varepsilon^2)$  and  $\|S_n^{\text{IEF},I}\|_2 \in \Theta(1)$ .

The singular case  $r_n = 0$  remains and must be handled explicitly. As done in [24], one practical solution is to add a damping factor  $\tau$  to the quotient, yielding

$$M_{\text{IEF},G} = \frac{1}{N} \sum_{n=1}^N \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau}, \quad g_n = J_n^\top r_n.$$

## 11 Output-Space Geometry

The original EWC derivation is tied to the true Fisher matrix, and therefore to the local KL geometry of probabilistic models [9]. From that perspective, one might conclude that EWC is theoretically tied to KL-type distances and therefore mismatched with settings that require different metrics. This concern is especially relevant in modern generative modeling, where Wasserstein-type structure appears explicitly in parts of the flow-matching and optimal-transport literature [29, 30]. However, our derivation of the empirical Fisher starting from **PL-CLOC** does not require compatibility with KL geometry. In fact, our derivation of  $F_{\text{EF}}$  sidesteps this problem by not relying on any output-space distance metric at all. This matches empirical results, where the empirical Fisher is useful in diffusion settings too [18]. In contrast, the preceding derivation and resulting form of the IEF is compatible with arbitrary output metrics. Concretely, the output-level derivation only requires a local metric  $G_n \succ 0$  on output displacements. In practice,  $G_n$  should be read as a quadratic surrogate of the output-space distance one wants to preserve local to the point  $f_{\theta_A^*}(a_n)$ . For the mathematical relation of  $M_{\text{IEF}}$  to  $G_n$ -based distance, see Section 5. The canonical Euclidean choice  $G_n = I$  recovers the algebraic form from [24].

For regression or NLL-based classification setups, this may already be the right local model. Other settings warrant other choices: if the outputs lie on a probability simplex, one may instead choose a local statistical geometry. If the outputs are elements of a distribution, a Wasserstein-type local geometry may be more appropriate; our experiments use a sliced-Wasserstein surrogate for this purpose [31].

## 12 Additional Theoretical Properties

We move further analysis of  $M_{\text{IEF}}$  into the appendix. Namely, we treat invariance to monotone relabelings of the loss and better hyperparameter interpretability in Appendix A, and analyze rank stability of the two matrices subject to a simple contamination model in Appendix B.

## 13 Improved Elastic Weight Consolidation

After the theoretical analysis of  $M_{\text{IEF}}$ , we turn to the remaining design choices needed for an implementation. First, in Section 10 we already identified the singular case  $r_n = 0$  and introduced damping with parameter  $\tau$  as the practical remedy. Second, using a full matrix is usually prohibitively expensive in memory and computation, so an approximation is necessary. [9] suggested a diagonal approximation, which we adopt, while noting the empirical evidence from diffusion continual learning that indicates that diagonal Fisher approximations can underperform in diffusion settings [18]. In addition, the contamination model of Appendix B suggests that the leading eigenspace of  $M_{\text{IEF}}$  can be more stable than that of  $F_{\text{EF}}$  in settings with atypical samples or samples that the model did not learn. Section 8 identified the empirical Fisher’s implicit sample-weighting profile. Since alternative aggregation weights can be useful in some continual-learning settings, we include an explicit weighting function; possible weighting mechanisms appear in FROMP and GSS [1, 23]. Therefore, let

$$h : \mathcal{D}_A \times \Theta \rightarrow \mathbb{R}_{\geq 0}, \quad a \mapsto h(a, \theta),$$

be the weighting function that assigns each sample its corresponding sample importance coefficient. To keep  $\lambda$  as the global regularization strength, normalize the coefficients as

$$\bar{h}(a_i, \theta) = \frac{h(a_i, \theta)}{\frac{1}{N_A} \sum_{j=1}^{N_A} h(a_j, \theta)},$$

In P-CLOC, this is simply absorbed into the penalty function. We recover the IEF from [24] with

$$\bar{h}(a_i, \theta) = 1,$$

The empirical Fisher corresponds instead to the unnormalized weighting profile

$$h(a_i, \theta) = w_i^G = r_i^\top G_i^{-1} r_i$$

when we use  $\tau = 0$ . With mean-normalized  $\bar{h}$ , this only differs by a global scale that can be absorbed into  $\lambda$ . This extension yields the final form of the importance matrix that we use:

$$M_{\text{IEF},G} = \frac{1}{N} \sum_{n=1}^N \bar{h}(a_n, \theta) \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau}, \quad g_n = J_n^\top r_n.$$

The free parameters of an implementation can be described more precisely as the four-tuple

$$(\lambda, \tau, \Gamma, h).$$

Here  $\lambda$  is the strength of the preservation constraint from CLOC,  $\tau$  controls damping for small-gradient samples,  $h$  maps samples to their importance coefficient and  $\Gamma$  specifies the output geometry. Concretely, we treat  $G_n$  as the samplewise metric induced by the chosen geometry model  $\Gamma$  at the old output.  $\Gamma$  maps an output to the positive-definite matrix that constitutes the local quadratic approximation of the distance:

$$\Gamma : \mathcal{Z} \rightarrow \mathbb{S}_{++}^m, \quad G_n = \Gamma(f_{\theta_A^*}(a_n)).$$

As motivated earlier, the canonical choice is to take  $\Gamma(\cdot) = I$ ,  $\tau \approx 0$  and  $h(\cdot, \cdot) = 1$ . Empirically,  $\tau = 10^{-2}$  is a practical default for the diagonal CIFAR-100 runs, and Appendix C shows that different values measurably affect forgetting. Conversely, the remaining parameter is then  $\lambda$  that intuitively controls the strength of the solution preservation constraint. For low-rank approximations of  $M_{\text{IEF}}$ , one also has to determine the rank  $k$ , turning this into a five-tuple. **Together, these derivations and implementation choices define an implementable mechanism derived from PL-CLOC: EWC arises from scalar loss preservation, and IEWC extends this mechanism through normalization, damping, sample weighting, and output-geometry selection. We call this mechanism Improved Elastic Weight Consolidation (IEWC).**

## 14 Empirical Evaluation

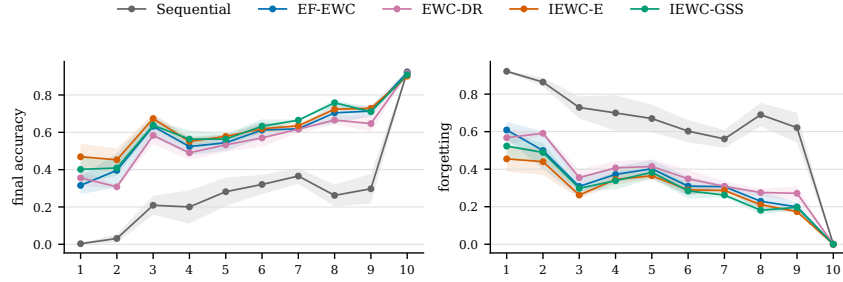
We next test whether the CLOC-derived regularizers produce useful continual-learning behavior in practical settings. Unless stated otherwise, all EWC matrices are diagonal. We write U-IEWC for the empirical, uniformly weighted IEF aggregation with  $h_n \equiv 1$ , and GSS-IEWC for a weighted variant that applies the gradient-diversity principle of GSS to the normalized IEWC directions [1]. The classification setting is ImageNet-R class-incremental learning with a pretrained ViT-B/16 and LoRA adapters, while the NLP setting is TRACE multiple-choice continual fine-tuning with Qwen2.5-3B and LoRA adapters. Thus, these two setups place the comparison in the recent line of parameter-efficient continual fine-tuning, including LoRA and recent continual LoRA methods [32–35]. We also evaluate non-LoRA fine-tuning of a pretrained CNN for VOC 15-5 semantic segmentation, and from-scratch training for long-horizon time-series forecasting and MNIST DDPM denoising. For each method and task type, we tune  $\lambda$  to optimize the average performance metric across distributions; Appendix C gives the full training and tuning protocol, including the selected  $\lambda$  and  $\tau$  values.

| Task type      | Metric                                | Sequential          | EF-EWC                                 | EWC-DR              | U-IEWC                                | GSS-IEWC                              |
|----------------|---------------------------------------|---------------------|----------------------------------------|---------------------|---------------------------------------|---------------------------------------|
| Classification | Final avg. accuracy $\uparrow$        | 0.2896 $\pm$ 0.0405 | 0.5981 $\pm$ 0.0337                    | 0.5687 $\pm$ 0.0263 | <b>0.6332 <math>\pm</math> 0.0232</b> | 0.6250 $\pm$ 0.0263                   |
| Classification | Avg. forgetting $\downarrow$          | 0.7066 $\pm$ 0.0466 | 0.3595 $\pm$ 0.0378                    | 0.3933 $\pm$ 0.0286 | <b>0.3142 <math>\pm</math> 0.0250</b> | 0.3285 $\pm$ 0.0278                   |
| NLP            | Final avg. score $\uparrow$           | 0.5760 $\pm$ 0.0027 | 0.5972 $\pm$ 0.0020                    | 0.5972 $\pm$ 0.0059 | 0.5938 $\pm$ 0.0034                   | <b>0.6272 <math>\pm</math> 0.0080</b> |
| NLP            | Avg. forgetting $\downarrow$          | 0.1139 $\pm$ 0.0111 | 0.0762 $\pm$ 0.0034                    | 0.0781 $\pm$ 0.0109 | 0.0820 $\pm$ 0.0122                   | <b>0.0046 <math>\pm</math> 0.0079</b> |
| Forecasting    | Final avg. MSE $\downarrow$           | 0.2733 $\pm$ 0.0065 | 0.2388 $\pm$ 0.0021                    | –                   | 0.2418 $\pm$ 0.0038                   | <b>0.2383 <math>\pm</math> 0.0007</b> |
| Forecasting    | Avg. MSE forgetting $\downarrow$      | 0.0535 $\pm$ 0.0081 | <b>0.0140 <math>\pm</math> 0.0020</b>  | –                   | 0.0191 $\pm$ 0.0047                   | 0.0147 $\pm$ 0.0021                   |
| Segmentation   | Final all mIoU $\uparrow$             | 0.0917 $\pm$ 0.0061 | 0.0946 $\pm$ 0.0040                    | 0.3080 $\pm$ 0.0307 | <b>0.4678 <math>\pm</math> 0.0568</b> | 0.4274 $\pm$ 0.0574                   |
| Segmentation   | Base-class forgetting $\downarrow$    | 0.7135 $\pm$ 0.0114 | 0.7131 $\pm$ 0.0166                    | 0.4471 $\pm$ 0.0447 | <b>0.2418 <math>\pm</math> 0.0965</b> | 0.2925 $\pm$ 0.0839                   |
| Diffusion      | Final avg. denoising MSE $\downarrow$ | 0.0676 $\pm$ 0.0029 | 0.0296 $\pm$ 0.0013                    | –                   | 0.0298 $\pm$ 0.0014                   | <b>0.0293 <math>\pm</math> 0.0013</b> |
| Diffusion      | Denoising-MSE forgetting $\downarrow$ | 0.0503 $\pm$ 0.0002 | <b>-0.0006 <math>\pm</math> 0.0020</b> | –                   | 0.0001 $\pm$ 0.0024                   | -0.0003 $\pm$ 0.0020                  |

**Table 1:** Main continual-learning results. The objective used for tuning is the final average performance across distributions; forgetting is reported separately as a retention measure. Higher is better for accuracy, score, and mIoU; lower is better for MSE and forgetting.

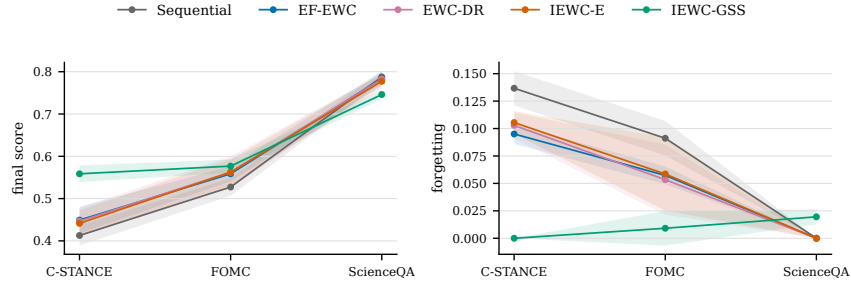
Overall, Table 1 separates the joint performance objective from the forgetting metric. U-IEWC gives the strongest ImageNet-R retention and average accuracy. In turn, GSS-IEWC gives the strongest TRACE result and the lowest average forecasting MSE, while EF-EWC is slightly better on forecasting forgetting. On VOC 15-5, U-IEWC gives the clearest gain over EF-EWC and EWC-DR. In DDPM

denoising, the tuned regularizers are close; GSS-IEWC has the lowest average denoising MSE, and EF-EWC has the smallest measured forgetting within the reported variance.



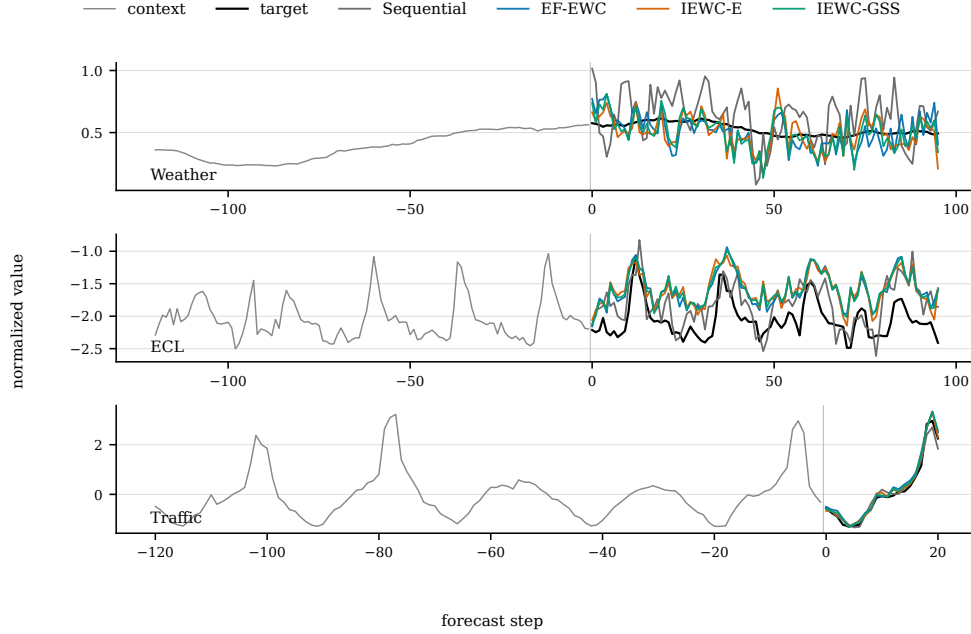
**Figure 2:** ImageNet-R distribution retention. The left panel shows the final accuracy on each distribution after the full sequence; the right panel shows the corresponding forgetting. U-IEWC and GSS-IEWC improve retention across most distributions relative to EF-EWC, while sequential training forgets the early distributions almost entirely.

At the distribution level, Figure 2 breaks the classification result down by distribution. The retained accuracy is not concentrated in a single late distribution: the IEWC variants reduce forgetting across the early and middle parts of the sequence, which is the regime where sequential fine-tuning fails most visibly.



**Figure 3:** TRACE distribution retention. The plot reports the final multiple-choice score and forgetting for C-STANCE, FOMC, and ScienceQA after sequential LoRA fine-tuning of Qwen2.5-3B. GSS-IEWC preserves the earlier NLP distributions while keeping competitive performance on the final distribution.

Similarly, the TRACE retention profile in Figure 3 explains the table-level gain of GSS-IEWC. The weighted aggregation mainly changes the stability side of the tradeoff: it reduces forgetting on C-STANCE and FOMC while leaving ScienceQA in the same range as the other regularized methods.



**Figure 4:** Forecasting rollout comparison. Each panel shows one held-out rollout from the corresponding forecasting distribution. The gray curve is the observed context, the black curve is the target horizon, and colored curves are method predictions after the full continual sequence.

For visual inspection and qualitative analysis, Figure 4 adds individual forecasts. We can see that forgetting significantly affects all methods, while GSS-IEWC retains the most target trajectory structure at the first task and stays close to EF-EWC on later tasks.

In addition, the output-geometry parameter  $\Gamma$  is evaluated separately in the DDPM setting. Replacing the Euclidean output metric with a sliced-Wasserstein surrogate lowers old-output sliced-Wasserstein drift relative to Euclidean U-IEWC ( $9.4409 \pm 7.2517$  versus  $14.1026 \pm 12.2220$ ), while keeping final average denoising MSE close to U-IEWC ( $0.0301 \pm 0.0012$  versus  $0.0298 \pm 0.0014$ ).

Finally, we tested FROMP-derived sample weighting [23] as a second alternative to the empirical weighting, with mixed results: on ImageNet-R, IEWC-FROMP reaches  $0.6365 \pm 0.0165$  final average accuracy and  $0.3068 \pm 0.0170$  forgetting, slightly ahead of GSS-IEWC; on VOC 15-5, it reaches only  $0.3351 \pm 0.0742$  final all-mIoU and  $0.4064 \pm 0.1122$  base-class forgetting, below U-IEWC and GSS-IEWC.

For reproducibility, Appendix C gives the experimental protocol, the programmatic  $\lambda$  search, a damping sensitivity curve, and the effective summand-norm distributions. Appendix D connects the contamination model of Appendix B to a controlled label-noise experiment and reports the resulting accuracy and importance-matrix stability.

## Code Release

We release an accompanying repository with a readily usable IEWC implementation. The `IEWCConfig` class encapsulates the four-tuple parametrization introduced above. The same configuration interface is used by the diagonal and low-rank IEWC paths and by the empirical scripts, including the FACIL-based CIFAR-100 checks and the larger ImageNet-R, TRACE, forecasting, segmentation, and diffusion runs. The repository is available at <https://github.com/Axym-Labs/iewc>.

## 15 Conclusion

We formulated continual learning as constrained optimization through CLOC and its population-setting, low-order variants. Within this framework, the empirical Fisher matrix used in EF-EWC is recovered exactly if we choose preserving the old loss value as constraint. This places the empirical Fisher matrix on solid theoretical grounding without relying on the true Fisher matrix or KL-type geometry. The same framework also instantiates preservation as distance to the old-task loss level set in output space; after first-order approximation, the resulting importance matrix is the improved empirical Fisher. A central result of our analysis is that the classification-specific failure mode addressed by EWC-DR is not an isolated pathology. Instead, it is one manifestation of a more general effect: EF-EWC scales old sample importance by the magnitude of the output loss gradient, suppressing samples that are already well fit. We show this explicitly in  $\Theta$ -Notation. The improved empirical Fisher removes this dependence and thereby yields a general solution that extends beyond classification to well-fitted old-task settings and arbitrary local output geometries. These design choices define a concrete continual-learning mechanism derived from first principles. In particular, this yields IEWC as a practical successor to diagonal Fisher regularization in EF-EWC. Across classification, NLP, forecasting, segmentation, and diffusion experiments, IEWC variants improve the average continual-learning objective or old-distribution retention over EF-EWC in the main settings; the sliced-Wasserstein diffusion experiment further illustrates that the output-geometry parameter can target a chosen notion of stability.

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## A Invariance to Loss Reparameterization

This appendix treats the undamped case  $\tau = 0$ .

The output-level feasible set

$$\mathcal{C}_n = \{z : \ell_n(z) = \ell_n(z_n^A)\}$$

depends on the level sets of  $\ell_n$ , not on the numerical labels assigned to those level sets. If we apply a local monotone scalar transformation

$$\tilde{\ell} = \phi(\ell), \quad \phi'(\ell) > 0,$$

then the output gradient rescales as

$$\tilde{r}_n = \phi'(\ell_n(z_n^A)) r_n.$$

For EF, the sample contribution changes by

$$J_n^\top \tilde{r}_n \tilde{r}_n^\top J_n = \phi'(\ell_n(z_n^A))^2 J_n^\top r_n r_n^\top J_n.$$

For IEF, the same scalar factor appears in numerator and denominator:



$$\frac{\tilde{r}_n \tilde{r}_n^\top}{\tilde{r}_n^\top G_n^{-1} \tilde{r}_n} = \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n}.$$

Thus IEF is invariant to local scalar relabelings of the constraint function. This includes loss rescaling, unit changes in regression losses, reduction conventions, and smooth monotone transformations that preserve the local level-set geometry. This invariance can also improve hyperparameter interpretability, as EF’s penalty scale depends on the loss magnitudes. IEF removes this dependence, while regularization strength and damping still require tuning.

## B Spectral Stability Under Contamination

The normalization difference between EF and IEF has consequences for their spectrum. This matters in practice because any diagonal or low-rank approximation is built from the leading part of the matrix. Consider for example [18]. To see how this difference can affect the leading eigenspace, consider the following simple contamination model: Contaminated samples, such as mislabeled or atypical data, can have large output-loss gradients because their signal aligns with directions that gradient descent learns only very slowly; in the spectral-bias / kernel-eigenmode framework, low-alignment or low-eigenvalue components can remain effectively unlearned over the training horizon [36, 37]. With probability  $1 - \varepsilon$ , a sample is clean and its normalized sensitivity direction contributes to a stable old-task preservation matrix  $\Sigma_{\text{clean}} = \mathbb{E}[aa^\top \mid a \in D_{A,\text{clean}}]$ . Assume also that clean weights are independent of these normalized directions, or are sufficiently concentrated around their clean mean. With probability  $\varepsilon$ , a sample is contaminated. Assume its direction is concentrated near a unit vector  $u$  outside that clean leading subspace, and that its output gradient scale is large,

$$w \approx W, \quad W \gg 1.$$

Under this model, the population version of IEF is

$$\mathbb{E}[M_{\text{IEF}}] = (1 - \varepsilon)\Sigma_{\text{clean}} + \varepsilon uu^\top$$

while the corresponding EF matrix becomes

$$\mathbb{E}[F_{\text{EF}}] = (1 - \varepsilon)\mathbb{E}[w \mid \text{clean}] \Sigma_{\text{clean}} + \varepsilon W uu^\top.$$

In terms of the spectrum, let  $\lambda_k^{\text{clean}}$  denote the  $k$ -th eigenvalue of the clean component. If

$$\varepsilon W \gtrsim (1 - \varepsilon)\lambda_k^{\text{clean}} \mathbb{E}[w \mid \text{clean}],$$

then the contaminated direction  $u$  will enter the leading eigenspace of  $F_{\text{EF}}$ . The insight is that the normalization can realistically affect which directions dominate the spectrum. In settings with noisy labels, hard examples, underfit outliers, or other rare but high-loss samples, we should expect the leading eigenspace of  $F_{\text{EF}}$  to be less stable than that of  $M_{\text{IEF}}$ , relevant to low-rank approximations thereof.

## C Experimental Details and Damping Sensitivity

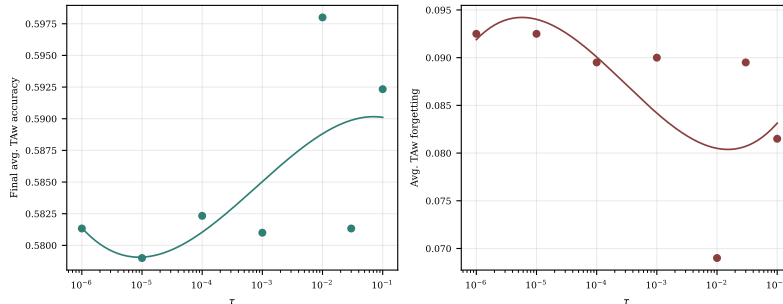
**Training and Tuning Protocol** Table 2 summarizes the paper-facing empirical runs. The tuning objective is the final average performance across all distributions in a sequence. For MSE-based tasks this means lower final average MSE; for accuracy, score, and mIoU it means higher final average performance. Forgetting is then reported as a separate retention metric rather than used as the primary tuning objective.

| Task type      | Dataset and sequence                      | and se-       | Model               | Pretrained | Training regime  | Selected regularization values                                                                                                                                    |
|----------------|-------------------------------------------|---------------|---------------------|------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classification | ImageNet-R, 20-class distributions        | ten distribu- | ViT-B/16            | Yes        | LoRA, rank 4     | EF: $\lambda = 333.333$<br>DR: $\lambda = 3$<br>U-IEWC: $\lambda = 33.333, \tau = 10^{-4}$<br>GSS-IEWC: $\lambda = 11.111, \tau = 10^{-2}$                        |
| NLP            | TRACE: STANCE FOMC → Sci-enceQA           | C- → Sci-     | Qwen2.5-3B          | Yes        | LoRA, rank 8     | EF: $\lambda = 1$<br>DR: $\lambda = 1$<br>U-IEWC: $\lambda = 0.111, \tau = 10^{-2}$<br>GSS-IEWC: $\lambda = 59049, \tau = 10^{-2}$                                |
| Forecasting    | Weather → ECL Traffic                     | → ECL →       | PatchTST            | No         | From scratch     | EF: $\lambda = 729$<br>U-IEWC: $\lambda = 9, \tau = 10^{-2}$<br>GSS-IEWC: $\lambda = 9, \tau = 10^{-2}$                                                           |
| Segmentation   | VOC 15-5, base classes then novel classes | base novel    | ConvNeXtV2-tiny FPN | Yes        | Full fine-tuning | EF: $\lambda = 0.111$<br>DR: $\lambda = 243$<br>U-IEWC: $\lambda = 270, \tau = 10^{-3}$<br>GSS-IEWC: $\lambda = 2430, \tau = 10^{-2}$                             |
| Diffusion      | MNIST DDPM, labels 0/1 → 2/3              | DDPM, la-     | Conditional DDPM    | No         | From scratch     | EF: $\lambda = 9 \cdot 10^5$<br>U-IEWC: $\lambda = 675, \tau = 10^{-3}$<br>GSS-IEWC: $\lambda = 225, \tau = 10^{-3}$<br>IEWC-SW: $\lambda = 10^6, \tau = 10^{-3}$ |

**Table 2:** Empirical training and tuning protocol for the main paper-facing runs. “DR” denotes EWC-DR. IEWC-SW is the sliced-Wasserstein output-geometry ablation discussed outside the main table.

The  $\lambda$  search is programmatic. For each regularized method, we evaluate a five-point multiplicative grid centered at a task-specific midpoint. The midpoint is either the selected value from a validated predecessor in the same task family or the best value from a short seed-0 scale probe. If the best point lies on a grid boundary, we extend the grid in that direction, with at most three such extensions. The selected  $\lambda$  is then rerun on the remaining seeds. This keeps the tuning objective fixed while avoiding ad hoc one-off extensions.

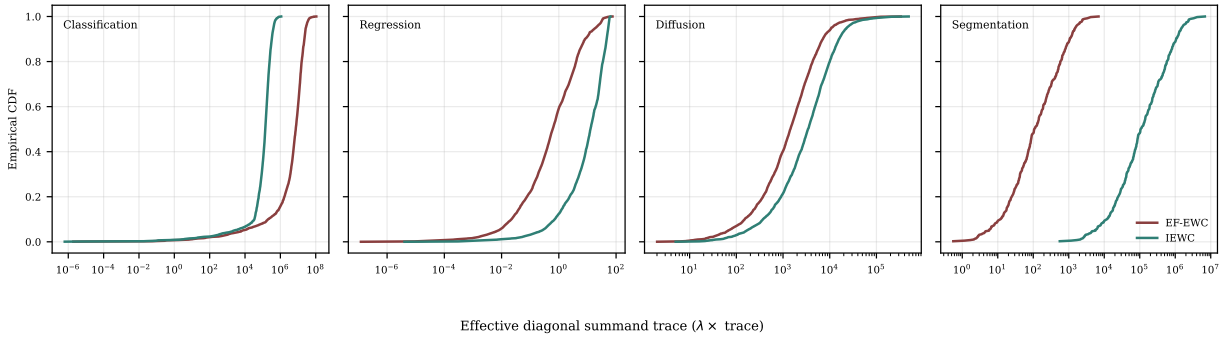
**Damping Sensitivity on CIFAR-100** Figure 5 reports a separate damping sensitivity experiment on CIFAR-100 with FACIL, the iCaRL class order, ResNet-32, and the first three distributions of the ten-distribution task-aware protocol. Each distribution is trained for 60 epochs. IEWC uses a diagonal matrix surrogate, 512 old-distribution samples for the importance estimate, and importances scale-matched to the corresponding EF estimate so that the EF regularization strength can be reused.



**Figure 5:** Tau sensitivity on CIFAR-100. The tested values are  $\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}\}$ . The damping parameter has a measurable effect on this ResNet/FACIL prefix protocol:  $\tau = 10^{-2}$  gives the highest final task-aware accuracy and lowest forgetting among the tested values, so we use it as a practical default before task-specific tuning.

| Method | $p = 0$             | $p = 0.1$           | $p = 0.25$          |
|--------|---------------------|---------------------|---------------------|
| EF-EWC | $0.5617 \pm 0.0062$ | $0.5132 \pm 0.0112$ | $0.2773 \pm 0.0320$ |
| EWC-DR | $0.5932 \pm 0.0296$ | $0.5310 \pm 0.0226$ | $0.4598 \pm 0.0095$ |
| IEWC   | $0.5978 \pm 0.0260$ | $0.5642 \pm 0.0215$ | $0.5322 \pm 0.0239$ |

**Effective Summand Norms** Figure 6 reports a separate scale diagnostic from the original diagonal-analysis suite: CIFAR-100 classification, controlled regression, MNIST DDPM denoising, and the earlier binary segmentation setup. The plot shows the effective scale of the diagonal summands after multiplication by the corresponding  $\lambda$ . It is not a sample-weight plot, because the unweighted U-IEWC mechanism uses  $h_n \equiv 1$ .



**Figure 6:** Effective diagonal summand norms. Each panel reports the empirical CDF of  $\lambda$  times the per-sample diagonal summand trace before averaging over old-distribution samples, with EF-EWC and IEWC shown on the same diagnostic task type.

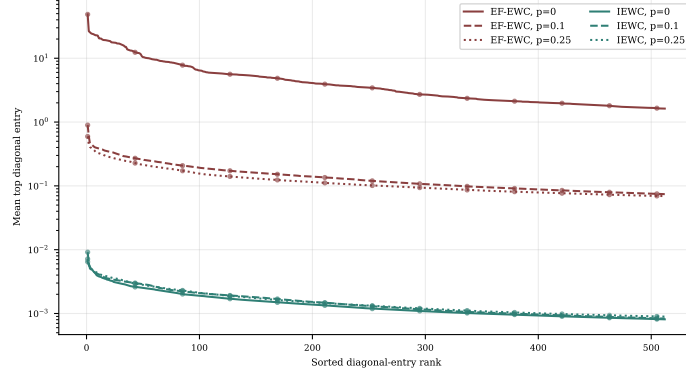
For EWC-DR, we additionally checked  $\lambda \in \{100, 300, 1000, 3000, 10000\}$  on the CIFAR-100 three-distribution prefix used for the damping plot. The best tested value on this prefix was  $\lambda = 100$ ; we use this only as a local parameter-sensitivity check for the earlier CIFAR-100 setup, not as the tuning protocol for the ImageNet-R results in Table 1.

## D Label-Noise Stress Test

The contamination model in Appendix B predicts that rare high-loss samples can dominate EF-EWC because their loss-gradient scales enter the empirical Fisher directly. Controlled label noise instantiates this mechanism: corrupted labels create atypical high-loss training samples, while clean task-aware test sets measure whether the stored preservation matrix still protects the original classification structure. We therefore add uniform label noise to the CIFAR-100 training data in FACIL and report final average task-aware accuracy. IEWC degrades much more slowly than EF-EWC as the label-noise rate increases.

To connect this performance effect to the matrix-level prediction of the contamination model, we also record the largest stored diagonal entries of the EF and IEWC matrices on the same CIFAR-100 prefix protocol. These are not low-rank approximations. For diagonal matrices, the relevant stability object is the high-importance coordinate tail, so the table uses all 512 exported entries per matrix and reports the stored-tail mass ratio, the normalized stored-tail profile change, and the log-scale RMSE of the sorted entries. EF-EWC changes much more under contamination than IEWC on the log-scale metric.

| Method | Noise | Entries | Stored-tail mass ratio | Stored-tail profile L1 | Stored-tail log-scale RMSE |
|--------|-------|---------|------------------------|------------------------|----------------------------|
| EF-EWC | 0.1   | 512     | 0.0288                 | 0.2610                 | 1.4660                     |
| EF-EWC | 0.25  | 512     | 0.0244                 | 0.2737                 | 1.5337                     |
| IEWC   | 0.1   | 512     | 1.0833                 | 0.0350                 | 0.0350                     |
| IEWC   | 0.25  | 512     | 1.1003                 | 0.0154                 | 0.0415                     |



**Figure 7:** Label-noise diagonal spectra. The plot visualizes the high-importance coordinate tail underlying the matrix-level comparison above; EF-EWC changes much more under contamination than IEWC on the log scale.